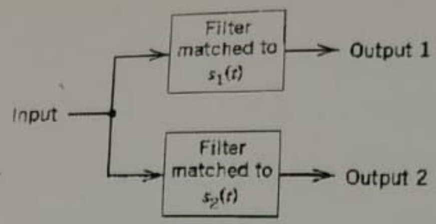


Figure 2

i. When the pulse $s_1(t)$ is applied to this two dimensional filter, the response of the lower matched filter is zero.

ii. When the pulse $s_2(t)$ is applied to this two dimensional filter, the response of the upper matched filter is zero.

4. The binary data stream $\{0011\}$ is applied to the input of a duobinary system. Assume that polar NRZ signalling/mapping scheme is used with initial mapped data as $\{a_{-1} = \hat{a}_{-1} = -1\}$.

   (a) Construct the duobinary coder output and corresponding receiver output, without a precoder.

   (b) Suppose that owing to error during transmission, the second sample at the receiver input $\{C_k\}$ is detected as zero. Construct the receiver output.

   (c) Repeat part(b), assuming the use of a precoder at the transmitter. Assume that initial precoder output is $\{d_{-1} = 0\}$

5. A pulse and its (raised cosine) spectral characteristic are shown in Figure 3. This pulse is used for transmitting digital information over a band-limited channel at a rate 1/T symbols/s. Find the roll-off factor and pulse transmission rate.

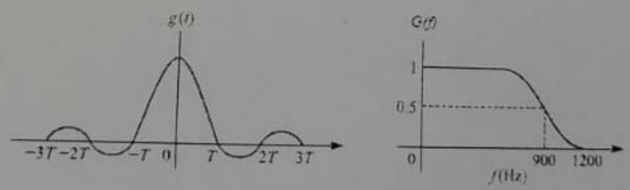


Figure 3

6. Binary PAM signal is used for transmitting data at a rate of 2400 bits/s over a channel having a frequency response

$$C(f) = \begin{cases} 1, & |f| \leq B/2 \\ 1/2, & B/2 < |f| < B \end{cases} \tag{3}$$

where $B = 1800$ Hz. AWGN is having PSD, $N_0/2 = 10^{-6}$ W/Hz. The average transmitter power is 1 mW. Determine the magnitude and frequency response characteristics of the optimum raised cosine transmitting and receiving filters for zero ISI.

Tutorial 2

ECC - 202 Digital Communication

February 24, 2026

1. A matched filter has a frequency response

$$H(f) = \frac{1 - e^{-j2\pi fT}}{j2\pi f} \tag{1}$$

   (a) Determine the impulse response $h(t)$ corresponding to $H(f)$.

   (b) Determine the signal waveform to which the filter characteristic is matched.

2. Consider the optimum detection of the sinusoidal signal

$$s(t) = sin(8\pi t/T),\ 0 \leqslant t \leqslant T \tag{2}$$

   (a) Determine correlator output, assuming a noiseless input.

   (b) Determine the corresponding matched filter output, assuming that the filter includes a delay T to make it causal.

   (c) Show that the outputs of correlator and matched filter are equal at $t = T$.

3. Figure 1 shows a pair of pulses that are orthogonal to each other over a time interval $[0, T]$.

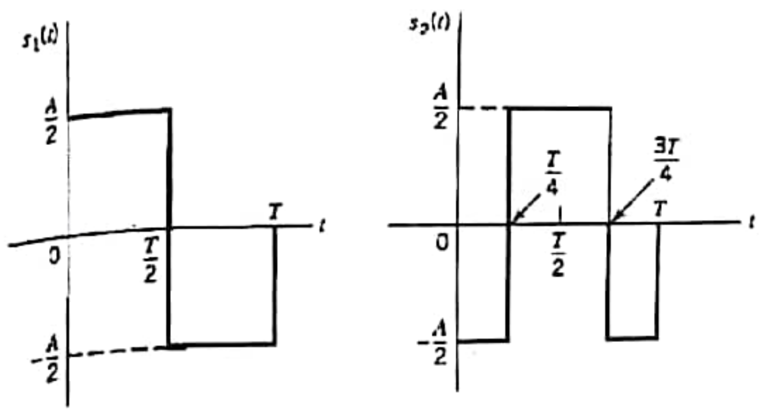


Figure 1

   (a) Determine the matched filter for the pulses $s_1(t)$ and $s_2(t)$ considered individually.

   (b) Form a two dimensional matched filter by connecting the two matched filters of part a in parallel, as shown in Figure 2. Hence demonstrate the following

Tutorial 1

ECC - 202 Digital Communication

February 11, 2026

1. Determine the Nyquist sampling rate for the signal $x(t) = \left(\frac{\sin(4000\pi t)}{\pi t}\right)^2$.

2. A baseband signal $m(t)$ has a bandwidth of 3 kHz. It is modulated as:

$$x(t) = m(t)cos(2\pi 20000t)$$

   Determine the Nyquist rate for $x(t)$.

3. A signal is given by

$$x(t) = cos(2\pi 5000t)$$

   It is passed through a nonlinear device whose characteristic is defined by $y(t) = x^2(t)$. Determine the Nyquist rate for $y(t)$.

4. Let $x(t)$ be a signal with Nyquist sampling rate ω_0. Also, let

$$y(t) = x(t)p(t-1) \tag{1}$$

   where

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT), \qquad T < \frac{2\pi}{\omega_0} \tag{2}$$

   Specify the constraints on the magnitude and phase of the frequency response of a filter that gives $x(t)$ as its output when $y(t)$ is the input.

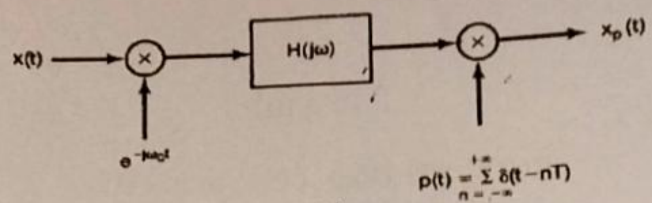
5. Another bandpass sampling procedure, used when $x(t)$ is real, consists of multiplying $x(t)$ by a complex-exponential and then sampling the product. The sampling system is shown in Figure 1. With $x(t)$ real and with $X(j\omega)$ nonzero only for $\omega_1 < |\omega| < \omega_2$, the frequency is chosen to be $\omega_0 = (1/2)(\omega_1 + \omega_2)$, and the lowpass filter $H(j\omega)$ has cutoff frequency $(1/2)(\omega_2 - \omega_1)$.

   (a) For $X(j\omega)$ as shown in Figure 1, sketch $X_p(j\omega)$.

   (b) Determine the maximum sampling period T such that $x(t)$ is recoverable from $x_p(t)$.

   (c) Determine a system to recover $x(t)$ from $x_p(t)$.

6. Consider a signal probability density function (pdf) given in Fig. 2. Calculate the signa power and find the signal-to-noise ratio due to quantization (SNRQ) in terms of L.



(a)

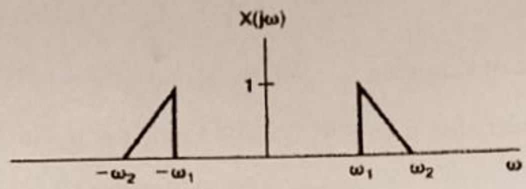


Figure 1

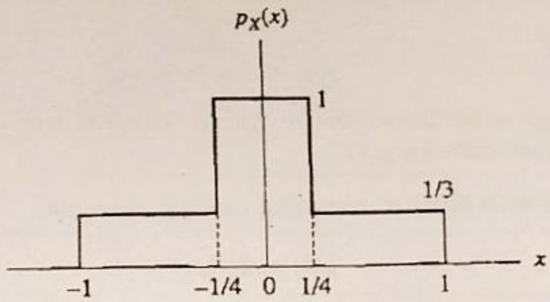


Figure 2

7. An analog signal has a voltage range of −5 V to +5 V. If this is quantised to 8 bits, determine the number of levels and the step size required for the quantised signal.

8. What is the maximum possible quantization error of the uniform quantizer used to quantize an input of amplitude 10 V peak to peak. Each level uses 2 bits for representation.

9. A TV signal having a bandwidth of 5 MHz is transmitted using binary PCM. If the number of quantization levels used in this PCM system is 512, determine the codeword length, transmission bandwidth, output bit rate and $(SNR)_q$

10. Consider a low-pass signal with a bandwidth of 3 KHz. A linear delta modulation system, with a step size $\Delta = 0.1$ V, is used to process this signal at a sampling rate 10 times the Nyquist rate.

    (a) Evaluate the maximum amplitude of a test sinusoidal signal of frequency 1 KHz, which can be processed by the system without slope-overload distortion.

    (b) For the specifications given in part 'a', evaluate the output SNR under prefiltered and postfiltered conditions.

INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DIGITAL COMMUNICATION

Quiz, Spring Semester, 2025

Time Allowed: ONE hour **Total Marks: 15**

---

1. A signal $x(t) = N\cos((200+N)\pi t) + (N+5)\cos((200-N)\pi t)$ undergoes ideal sampling at a frequency of 150 samples per sec where N represents the last three digits of your enrollment number. Further, the sampled signal is filtered using a unit gain low pass filter (LPF) with Nyquist cut-off frequency.

   (a) Draw the amplitude spectra of the sampled signal. (1 mark)

   (b) Determine the frequency components present in the output of the LPF. (1 mark)

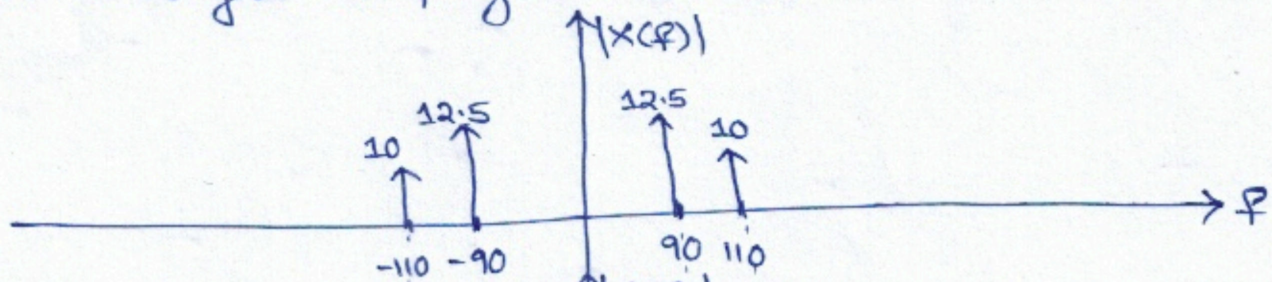
   (c) Write the time-domain expression for the output of the LPF. (2 marks)

Considering $N=20$, $x(t) = 20\cos(220\pi t) + 25\cos(180\pi t)$

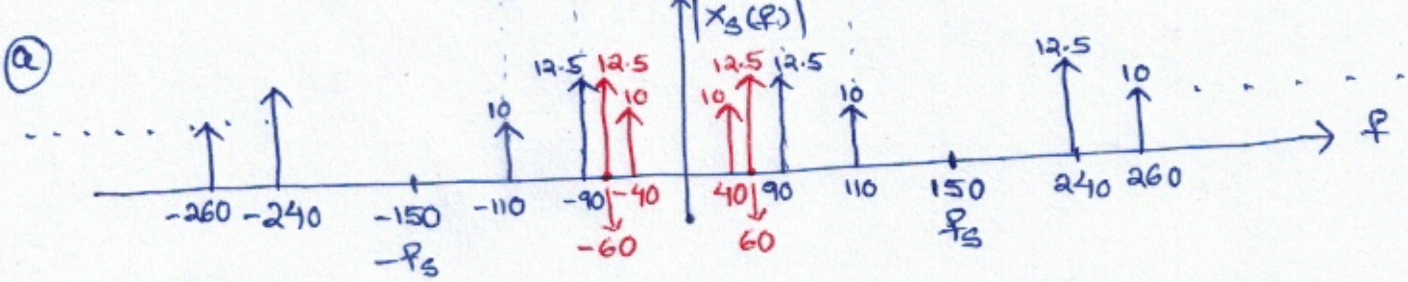
$$= 20\cos(2\pi \cdot 110 \cdot t) + 25\cos(2\pi \cdot 90 \cdot t)$$

$$\Rightarrow X(f) = 10\left[\delta(f+110) + \delta(f-110)\right] + 12.5\left[\delta(f+90) + \delta(f-90)\right]$$

$x(t)$ undergoes sampling at $f_s = 150$ Hz.



(a)



(b) Frequency components present in the output of LPF are

$$\underbrace{40\text{ Hz},\ 60\text{ Hz}}_{\text{Aliased frequencies}},\ \underbrace{90\text{ Hz},\ 110\text{ Hz}}_{\text{Original frequencies}}$$

(c) Output of LPF is $x_r(t) = 20\cos(2\pi \cdot 110 t) + 25\cos(2\pi \cdot 90 t) + 20\cos(2\pi \cdot 40 t) + 25\cos(2\pi \cdot 60 t)$

2. Consider a signal whose probability density function is shown in Figure 1. Consider uniform

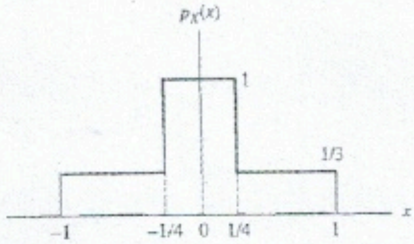


Figure 1

quantization with $L = 2^N$ where N represents the last three digits of your enrollment number. Determine the following:

(a) Quantization error variance (2 marks)

(b) Average signal to quantization noise ratio (1 mark)

(a) Quantization error variance :

$$\sigma_Q^2 = \frac{1}{3L^2} = \frac{1}{3(2^N)^2} = \frac{1}{3(2^{2N})}$$

(b) Average Signal Power :

$$\sigma_x^2 = 2\left[1 \times \int_0^{1/4} x^2\,dx + \frac{1}{3} \times \int_{1/4}^{1} x^2\,dx\right]$$

$$= 2\left[\frac{1}{192} + \frac{21}{192}\right] = 2 \times \frac{22}{192} = 0.2291$$

$$(SNR)_Q = \frac{0.2291}{1/(3 \times 2^{2N})} = 0.6875 \times 2^{2N}$$

3. A μ-law compander is defined by

$$y = \pm\frac{ln(1+\mu|x|)}{ln(1+\mu)}, \quad \text{for } |x| < 1$$

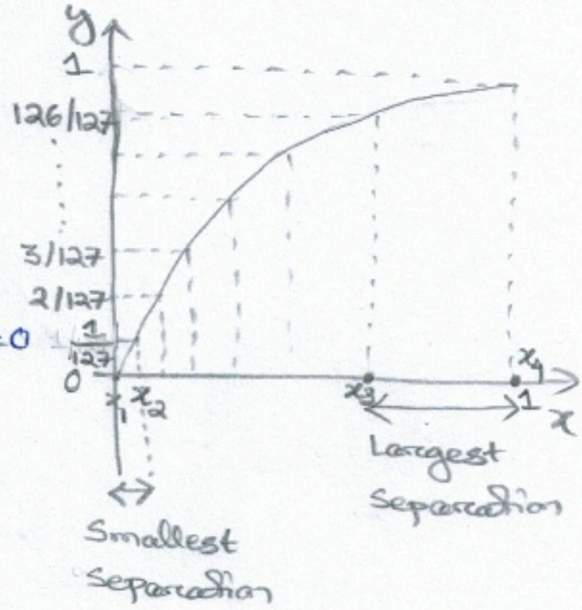
where x is the input and y is the output. The + sign is used when x is positive, and the - sign is used when x is negative. Consider the peak value of input is 10 volts, and the number of bits available for quantization are 8. If $\mu = N$ where N represents the last three digits of your enrollment number, determine the following:

(a) Smallest separation between levels (2 marks)

(b) Largest separation between levels (2 marks)

Considering the normalized input $|x| < 1$ i.e. $-1 < x < 1$, the Y-axis is uniformly quantized with step size

$$\frac{1}{\left(\frac{2^8}{2}\right)-1} = \frac{1}{127}$$



Let $N = 255 \Rightarrow \mu = N = 255$

ⓐ Smallest separation occurs near to $x=0$ i.e. between $y_1 = 0$ and $y_2 = \frac{1}{127}$

$$\therefore \frac{\ln(1+255x_1)}{\ln(1+255)} = 0 \Rightarrow x_1 = 0$$

$$\frac{\ln(1+255x_2)}{\ln(1+255)} = \frac{1}{127} \Rightarrow x_2 = 1.75\times10^{-4}$$

$\therefore$ Smallest separation $= 10(x_2 - x_1) = 1.75\times10^{-3}$

ⓑ Largest separation occurs on x-axis between the levels $y_3 = \frac{126}{127}$ and $y_4 = 1$. Then, $\frac{\ln(1+255x_3)}{\ln(1+255)} = \frac{126}{127} \Rightarrow x_3 = 0.957$

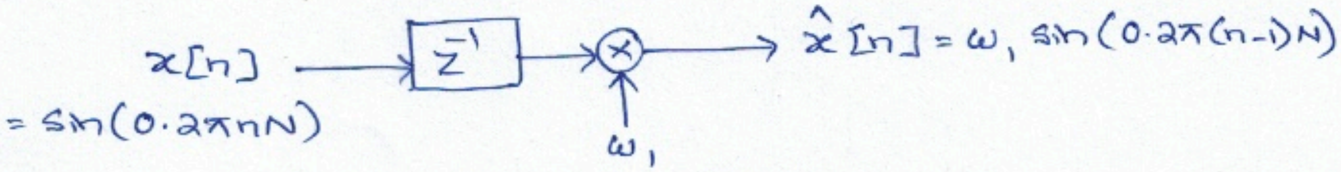
Similarly, $\frac{\ln(1+255x_4)}{\ln(1+255)} = 1 \Rightarrow x_4 = 1$

$\therefore$ Largest separation $= 10(x_4 - x_3) = 10(1 - 0.957) = 0.43$

4. The input signal to a one-tap linear predictor is given by $x[n] = \sin(0.2\pi n N)$ where N represents the last three digits of your enrollment number. Determine the following:

(a) Tap weight for which the prediction error variance is minimized (1 mark)

(b) Minimum prediction error variance (1 mark)



Prediction error $e[n] = x[n] - \hat{x}[n]$

(a) Prediction error variance $J = E[e^2[n]]$

$$\Rightarrow J = E[x^2[n]] + E[\hat{x}^2[n]] - 2E[x[n]\hat{x}[n]]$$

$$\Rightarrow J = E[\sin^2(0.2\pi n N)] + \omega_1^2 E[\sin^2(0.2\pi(n-1)N] - 2\omega_1 E[\sin(0.2\pi n N)\sin(0.2\pi(n-1)N)]$$

$$= \frac{1}{2} + \omega_1^2 \times \frac{1}{2} + \omega_1 E[-2\sin(0.2\pi n N)\sin(0.2\pi(n-1)N)]$$

$$= \frac{1}{2} + \frac{1}{2}\omega_1^2 + \omega_1 E[\cos(0.4\pi n N - 0.2\pi N) - \cos(0.2\pi N)]$$

$$= \frac{1}{2} + \frac{1}{2}\omega_1^2 + \omega_1 \times 0 - \omega_1 \cos(0.2\pi N)$$

$$\frac{\partial J}{\partial \omega_1} = \omega_1 - \cos(0.2\pi N) = 0 \Rightarrow \omega_1 = \cos(0.2\pi N)$$

(b) $$J_{min} = \frac{1}{2} + \frac{1}{2}\cos^2(0.2\pi N) - \cos^2(0.2\pi N)$$

$$= \frac{1}{2} - \frac{1}{2}\cos^2(0.2\pi N) = \frac{1}{2}\sin^2(0.2\pi N)$$

5. A message signal $x(t) = 3\ \sin(Nt) + 4\ \sin(2Nt) + 4\ \sin(1.5Nt)$ is encoded using a delta modulator (DM) where N represents the last three digits of your enrollment number. Assuming no slope-overload distortion, determine the following:

(a) Quantization step-size (1 mark)

(b) Maximum output signal-to-quantization noise ratio (1 mark)

$$x(t) = 3\sin(Nt) + 4\sin(2Nt) + 4\sin(1.5Nt)$$

(a) $$\frac{dx(t)}{dt} = 3N\cos(Nt) + 8N\cos(2Nt) + 6N\cos(1.5Nt)$$

$$\left|\frac{dx(t)}{dt}\right|_{max} = 3N + 8N + 6N = 17N$$

To avoid slope-overload distortion, $\frac{\Delta}{T_s} \geqslant \left|\frac{dx(t)}{dt}\right|_{max}$

$$\therefore \frac{\Delta}{T_s} \geqslant 17N \Rightarrow \Delta \geqslant 17NT_s \Rightarrow \Delta \geqslant \frac{17N}{f_s}$$

Maximum frequency in the signal $\omega_m = 2N \Rightarrow f_m = \frac{2N}{2\pi} = \frac{N}{\pi}$

Assuming that the signal is sampled at Nyquist rate, $f_s = \frac{2N}{\pi}$

$$\therefore \Delta \geqslant \frac{17N}{2N/\pi} \Rightarrow \Delta \geqslant 8.5\pi$$

(b) Prefiltered signal-to-quantization noise ratio:

$$\text{Maximum}(SNR)_Q = \frac{\text{Signal Power}}{\text{Quant Noise Power}} = \frac{\text{Signal Power}}{\Delta^2/3} = \frac{\frac{3^2}{2} + \frac{4^2}{2} + \frac{4^2}{2}}{(8.5\pi)^2/3}$$

$$= 0.0862$$

INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DIGITAL COMMUNICATION

Quiz-2, Spring Semester, 2025

Time Allowed: ONE hour **Total Marks: 15**

1. A voice signal in the range 300 to 3300 Hz is sampled at 8000 samples/sec. We may transmit these samples directly as PAM pulses or we may first convert each sample to a pulse-code modulation (PCM) format and use binary PCM waveforms for transmission.

   (a) What is the minimum system bandwidth required for the detection of PAM with no ISI and with a filter roll-off characteristic of $\alpha = N \times 10^{-4}$ where N represents the last four digits of your enrollment number? (1 mark)

   (b) Using the same filter roll-off characteristic, what is the minimum bandwidth required for the detection of binary PCM waveforms if the samples are quantized to eight levels? (1 mark)

Voice signal range is 300 Hz to 3300 Hz

Sampling frequency $f_s = 8000$ samples/sec $= R_s$

(a) System bandwidth $B_T = W(1+\alpha)$

For PAM, $W = \frac{R_s}{2} = \frac{8000}{2} = 4000$

$B_T = 4000(1+\alpha) = 4000(1+0.0001N)$

(b) PCM using 8-level quantization

Bit rate $= R_b = 8000 \times \log_2 8 = 24000$ bits/sec

$B_T = \frac{R_b}{2}(1+\alpha) = 12000(1+0.0001N)$

(Refer to Q1 of Tutorial-4)

2. For an AWGN channel, prove that the outputs of correlators in a correlator receiver are statistically independent of each other. (3 marks)

* Derived in the lecture class

* Pages 319 – 320 of Reference Book i.e. "Communication Systems" by Simon Haykin, 4th edition

3. Assume that N represents the last four digits of your enrollment number. A binary communication system transmits bits, x_n, where $x_n \in \{-1, +1\}$ with equal probability. The overall pulse response $p(t)$ is

$$p_k = p(kT) = \begin{cases} 1, & k = 0 \\ N \times 10^{-4}, & k = 1, 2 \\ 0, & \text{elsewhere} \end{cases}$$

(a) Determine the possible values for the ISI and their probabilities. (2 marks)

(b) Derive an expression for the probability of error. (2 marks)

The receiver output samples are

$$y_i = \underbrace{x_i p(0)}_{\text{desired component}} + \underbrace{\sum_{j \neq i} x_j\, p[(i-j)T_b]}_{\text{ISI component}} + \underbrace{n_i}_{\text{noise component}}$$

In this case, $y_i = x_i p(0) + \underbrace{x_{i-1} p(T_b) + x_{i-2} p(2T_b)}_{\text{ISI}} + n_i$

$$\Rightarrow y_i = x_i + 0.0001N\,(x_{i-1} + x_{i-2}) + n_i$$

(a)

x_{i-1}	x_{i-2}	ISI	Probability
-1	-1	$-0.0002N$	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$
-1	$+1$	0	$\frac{1}{4}$
$+1$	-1	0	$\frac{1}{4}$
$+1$	$+1$	$0.0002N$	$1/4$

Therefore, $\text{ISI} = \begin{cases} -0.0002N & \text{with prob. } 1/4 \\ 0 & \text{with prob. } 1/2 \\ 0.0002N & \text{with prob. } 1/4 \end{cases}$

If $x_i = +1$, then $y_i = 1 + \text{ISI} + n_i$

(b) Let us assume that variance of n_i is σ_n^2. Then, the prob. of error $P_e = \frac{1}{4} Q\left(\frac{1-0.0002N}{\sigma_n}\right) + \frac{1}{2} Q\left(\frac{1}{\sigma_n}\right) + \frac{1}{4} Q\left(\frac{1+0.0002N}{\sigma_n}\right)$

(Refer to Q2 of Tutorial 5) * You can also write in terms of erfc() if it is convenient to you.

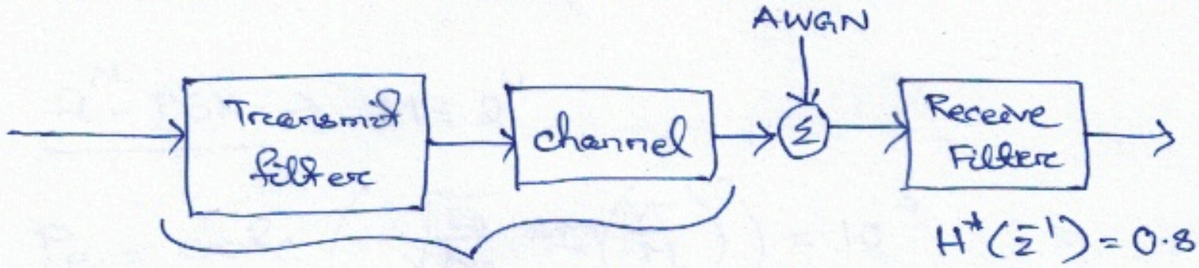
4. Consider the transfer function for the combined system of transmit filter and channel as

$$H(z) = 0.8 + (N \times 10^{-4})\, z^{-1}$$

where N represents the last four digits of your enrollment number. Assuming that the one-sided noise power spectral density is 0.1 W/Hz.

(a) Determine the tap coefficients of a three-tap zero-forcing equalizer. (1 mark)

(b) Determine the tap coefficients of a three-tap MMSE based equalizer. (2 marks)



$$H^*(z^{-1}) = 0.8 + (0.0001N)z$$

$$H(z) = 0.8 + (0.0001N)z^{-1}$$

$$X(z) = H(z)\,H^*(z^{-1}) = [0.8 + (0.0001N)z^{-1}][0.8 + (0.0001N)z]$$

$$= 0.64 + 0.0001N(z^{-1} + z) + (0.0001N)^2$$

$$= 0.64 + (0.0001N)^2 + 0.0001N(z^{-1} + z)$$

Let $N = 6100 \Rightarrow X(z) = 0.64 + (0.61)^2 + 0.61(z^{-1} + z)$

$$= 1.0121 + 0.61(z^{-1} + z)$$

(a)
$$\begin{bmatrix} 1.0121 & 0.61 & 0 \\ 0.61 & 1.0121 & 0.61 \\ 0 & 0.61 & 1.0121 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} -2.1774 \\ 3.6128 \\ -2.1774 \end{bmatrix}$$

(Refer to Q4 of Tutorial-4)

(b) One-sided noise PSD $= 0.1$ W/Hz

Two-sided " " $= \frac{0.1}{2} = 0.05$ W/Hz

$$\begin{bmatrix} 1.0121+0.05 & 0.61 & 0 \\ 0.61 & 1.0121+0.05 & 0.61 \\ 0 & 0.61 & 1.0121+0.05 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 0.61 \\ 0.8 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} -0.1402 \\ 1.2442 \\ -0.7146 \end{bmatrix}$$

5. Among two bandpass data transmission systems, one system uses 2^N-PSK, and the other system uses 2^N-QAM where N represents the last four digits of your enrollment number. The required average probability of symbol error for both systems is equal to 10^{-3}.

(a) Determine E_s/N_0 in terms of erfc() required for the PSK system. (1 mark)

(b) Determine the average E_s/N_0 in terms of erfc() required for QAM system. (1 mark)

(c) Compare the energy-to-noise PSD ratio requirements of these two systems. (1 mark)

(a) 2^N-PSK $\rightarrow$ $M = 2^N$

$$P_e = \text{erfc}\left(\sqrt{\frac{E_s}{N_0}}\sin\left(\frac{\pi}{M}\right)\right) = 10^{-3}$$

$$\Rightarrow \frac{E_s}{N_0} = \left(\frac{\text{erfc}^{-1}(10^{-3})}{\sin(\pi/M)}\right)^2$$

(b) 2^N-QAM $\rightarrow$

$$P_e = 2\left(1 - \frac{1}{\sqrt{M}}\right)\text{erfc}\left(\sqrt{\frac{3E_{av}}{2(M-1)N_0}}\right) = 10^{-3}$$

$$\Rightarrow \frac{E_{av}}{N_0} = \left[\text{erfc}^{-1}\left(\frac{10^{-3}}{2\left(1 - \frac{1}{\sqrt{M}}\right)}\right)\right]^2 \times \frac{2(M-1)}{3}$$

(c) 2^N-PSK demands more symbol energy than 2^N-QAM for the same P_e.

Problem 6.16 of Reference Book i.e. "Communication Systems" by Simon Haykin, 4th edition

INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DIGITAL COMMUNICATIONS

Mid-Term Examination, Spring Semester, Date: 10 March 2025

Time Allowed: 1.5 hours **Total Marks: 30**

---

1. Assume that the input message sequence is $\{1.2, -0.2, -0.5, 0.4, 0.89, 1.3\}$. Determine the **quantized sequence** and the corresponding **binary bitstream**, if the input sequence is quantized in the range (-1.5, 1.5) with 4 levels using a

   (a) uniform quantizer (2 marks)

   (b) μ-law quantizer where $\mu = 9$ (3 marks)

   * **NOTE:** A μ-law compressor with normalized input x and normalized output y is defined by

$$y = \pm \frac{ln(1 + \mu|x|)}{ln(1 + \mu)}$$

2. A message signal $m(t) = (4/\pi)\sin(2\pi 500t)$ is sampled, and delta modulation is used for digital representation and transmission. In order to meet the requirements that there be no slope overload noise and that the granular noise can not exceed 0.1 volts,

   (a) determine the maximum step size (1 mark)

   (b) determine the minimum sampling frequency (2 marks)

3. Increase in signal-to-quantization noise ratio with bit rate is much more dramatic for pulse code modulation (PCM) than that for delta modulation (DM). Justify this statement using an example. (3 marks)

4. Prove mathematically that the minimum mean square error achieved by a linear optimum predictor (Wiener filter) is always less than the variance of the input signal that is being predicted. (3 marks)

5. Given a multiplexer with sampling frequency of 8 KHz that accommodates six input signals having bandwidths of 8 KHz, 3.5 KHz, 2 KHz, 1.8 KHz, 1.5 KHz and 1.2 KHz. **In the multiplexer output, it is ensured that successive samples of each input signal are equispaced in time.**

   (a) Design the multiplexer involving submultiplexers and frame marker. (4 marks)

   (b) Determine the resulting bandwidth of the multiplexer. (1 mark)

6. A band-limited signal $y(t)$ is impulse sampled with a sampling time period of T_s. If $Y(\omega) = 0$ for $|\omega| \geq \frac{\pi}{T_s}$, determine a reconstruction filter $h(t)$ such that (2 marks)

$$\frac{dy(t)}{d(t)} = \sum_{n=-\infty}^{\infty} y(nT_s)h(t - nT_s)$$

7. Consider a signal s(t) as shown in Figure 1. A filter $h(t)$ is designed by matching it to s(t). With s(t) as input, the output of $h(t)$ is sampled at 800 Hz.

0.5

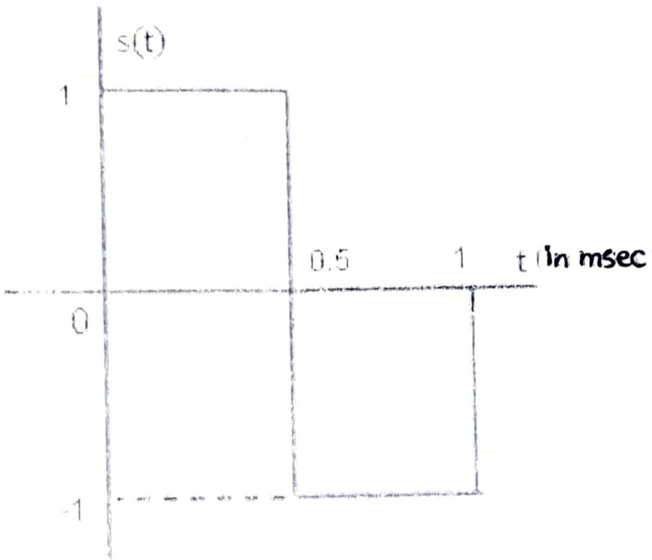


Figure 1

(a) Determine $h(t)$. (1 mark)

(b) Sketch $h(t)$. (1 mark)

(c) Determine the output of $h(t)$. (2 marks)

(d) Sketch the output of $h(t)$. (1 mark)

8. (a) Define the frequency-domain Nyquist criterion for zero inter-symbol interference. (1 mark)

(b) Consider a pulse of duration T_b whose spectrum is shown in Figure 2. Justify with explanation whether this spectrum will satisfy the Nyquist criterion for zero inter-symbol interference for

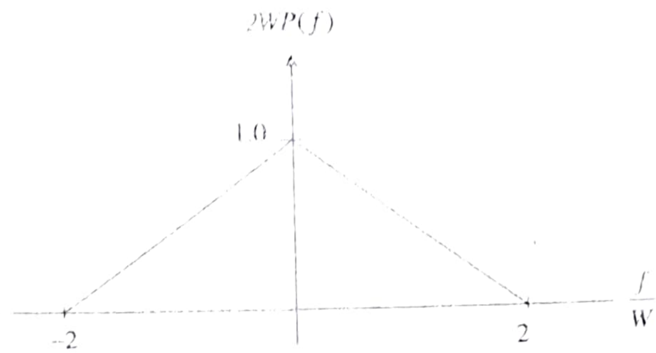


Figure 2

(i) $T_b < 1/(2W)$ (1 mark)

(ii) $T_b = 1/(2W)$ (1 mark)

(iii) $T_b > 1/(2W)$ (1 mark)

NOTE: Answers without justification will not be considered

INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DIGITAL COMMUNICATIONS

End-Term Examination, Spring Semester, 2025

Time Allowed: 3 hours **Total Marks: 40**

NOTE: Attempt ALL questions. All parts of a question must be answered in continuation. Make justifiable assumptions, if necessary, and clearly state these.

1. An M-ary PAM system transmits binary data at a rate of 4×10^6 bits/sec over a bandlimited AWGN channel with two-sided noise power spectral density $N_0/2$ watts/Hz using the raised cosine roll-off characteristic shown in the Figure 1.

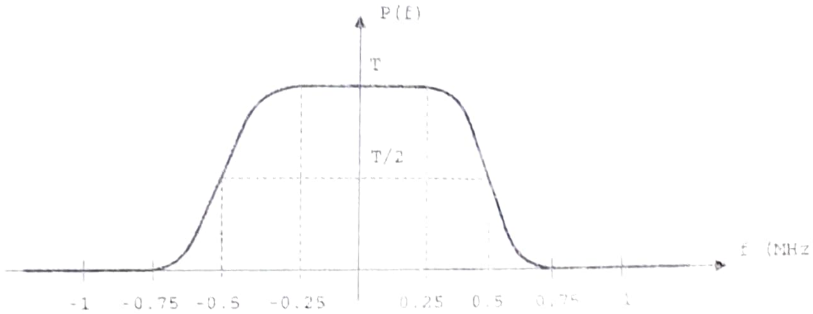


Figure 1

(a) Determine the roll-off factor and the value of M for the PAM system. (2 marks)

(b) Determine the minimum average bit energy-to-noise power spectral density ratio E_b/N_0 (in dB) required to achieve a bit error probability less than 10^{-3}. (2 marks)

2. The overall pulse response $P(f)$ of a duobinary signalling scheme is generated by the block shown in Figure 2, where $H_N(f)$ is an ideal Nyquist (low-pass) filter with cut-off frequency of $W = 1/(2T)$ Hz.

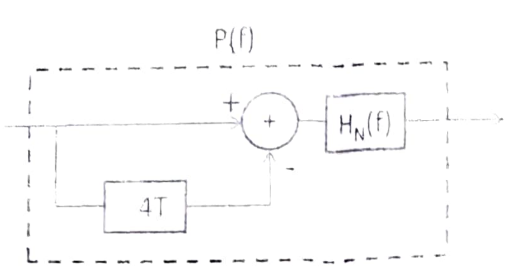


Figure 2

(a) Determine $P(f)$, and sketch the magnitude and phase response. (3 marks)

(b) Determine the time-domain pulse $p(t)$ and sketch it. (2 marks)

(c) Using $P(f)$ from part (a), determine the frequency response of the corresponding transmit and receive matched filters. (2 marks)

3. Consider a two level PAM system with transmitted levels 1 and −1. The channel through which the data is transmitted introduces intersymbol interference and corresponding received signal is given by

$$y_k = I_k + 0.2 I_{k-1} + n_k$$

where y_k, I_k and n_k represent the received sample, input sample, and AWGN sample at k^{th} time instant, respectively. The received samples at $k = 1$ to 4 are $\{0.8, -0.2, 0.4, 1\}$. Use the maximum likelihood sequence estimation based on Viterbi algorithm to determine the surviving sequence. (4 marks)

4. A binary communication system uses the following pair of waveforms for binary signaling

$$s_1(t) = u_T(t)$$
$$s_2(t) = \cos(2\pi t/T)\, u_T(t)$$

where $u_T(t) = u(t) - u(t-T)$, and $u(t)$ is the unit step function.

(a) Find a complete set of basis functions to represent these two signals. (2 marks)

(b) Find the signal vectors $\mathbf{s}_1$ and $\mathbf{s}_2$, and plot them in the signal space. (1 mark)

(c) Assume that $s_1(t)$ and $s_2(t)$ are used with equal probability in an AWGN channel with two-sided noise power spectral density $N_0/2$ watts/Hz. Find the probability of bit error in terms of average bit energy-to-noise PSD ratio E_b/N_0. (2 marks)

5. Consider the constellation diagram shown in Figure 3 used for data transmission over an AWGN channel with two-sided noise power spectral density $N_0/2$ watts/Hz. Assume that the symbols are grey encoded.

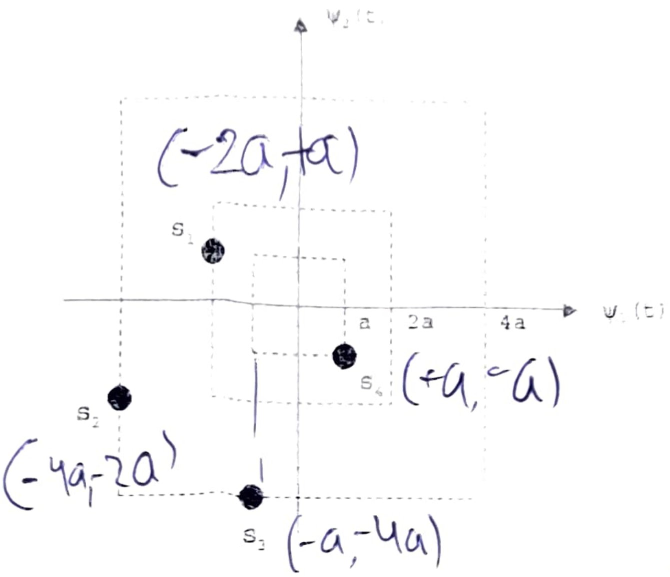


Figure 3

(a) Find the probability of bit error in terms of average bit energy. (2 marks)

Compare the result in (a) with the corresponding expression for a conventional QPSK system, and determine the loss or gain in SNR in dB if the probability of bit error is kept same for both the systems. (2 marks)

6. (a) Justify mathematically, why the minimum separation between the two frequencies used in binary FSK for ensuring orthogonality does not guarantee continuous phase at bit transitions. (2 marks)

   (b) Define the expression for a general continuous phase FSK signal. (1 mark)

   (c) Justify mathematically, how minimum shift keying (MSK) maintains minimum separation between two frequencies in addition to continuous phase at bit transitions. (2 marks)

   (d) Draw the phase state trellis for the MSK signals for the time duration $[0, 3T]$ with initial phase of $\pi/2$. (1 mark)

7. Assume that X is a uniformly distributed random variable in the interval $[\theta\ 1]$ whose probability density function is defined as

$$f_\theta(x) = \begin{cases} \dfrac{1}{1-\theta} & ; \theta \leq x \leq 1 \\ 0 & ; \text{otherwise} \end{cases}$$

The samples $\{x_1, \ldots, x_N\}$ were drawn independently from this distribution. Find the maximum likelihood estimate of θ. (2 marks)

8. Assume a mid-rise type symmetric uniform quantizer having L number of quantization levels.

   (a) Derive the output signal-to-noise ratio if the input X is uniformly distributed over $[-0.5, 0.5)$ and quantization error is uniformly distributed over $[-\Delta/2, \Delta/2)$ where Δ represents the step size of the quantizer. (1 mark)

   (b) Suppose the quantizer in (a) is changed to another quantizer by uniformly distributing X over $[-1, 1)$. The characteristic $f(X)$ of the new quantizer is defined as

$$f(X) = \begin{cases} -0.5 & ; \text{for } -1 \leq X < -0.5 \\ \text{Uniform Quantizer with step size } \Delta = 1/L & ; \text{for } -0.5 \leq X < 0.5 \\ 0.5 & ; \text{for } \ 0.5 \leq X < 1 \end{cases}$$

   Determine the variance of the quantization error. (3 marks)

   (c) What will be the resulting output SNR for the new quantizer in (b)? (1 mark)

   (d) Is it possible to increase SNR beyond the value obtained in (c) by increasing the number of bits required per sample? Justify your answer. (1 mark)

9. Consider the continuous signal

$$x(t) = 3\cos(50\pi t) + 10\sin(300\pi t) - \cos(100\pi t)$$

   (a) Assume that an ideal low pass filter is used for reconstruction process. Is it possible to reconstruct $x(t)$ exactly if $x(t)$ is sampled at the Nyquist rate? Justify your answer with sufficient reasoning. (1 mark)

   (b) Assuming that the sampling rate is fixed at the Nyquist rate due to hardware limitation, provide a solution for exact reconstruction of $x(t)$. (1 mark)

Complementary Error Function Table

x	erfc(x)	x	erfc(x)	x	erfc(x)	x	erfc(x)	x	erfc(x)	x	erfc(x)	x	erfc(x)
0	1.000000	0.5	0.479500	1	0.157299	1.5	0.033895	2	0.004678	2.5	0.000407	3	0.00002209
0.01	0.988717	0.51	0.470756	1.01	0.153190	1.51	0.032723	2.01	0.004475	2.51	0.000386	3.01	0.00002074
0.02	0.977435	0.52	0.462101	1.02	0.149162	1.52	0.031587	2.02	0.004281	2.52	0.000365	3.02	0.00001947
0.03	0.966159	0.53	0.453536	1.03	0.145216	1.53	0.030484	2.03	0.004094	2.53	0.000346	3.03	0.00001827
0.04	0.954889	0.54	0.445061	1.04	0.141350	1.54	0.029414	2.04	0.003914	2.54	0.000328	3.04	0.00001714
0.05	0.943628	0.55	0.436677	1.05	0.137564	1.55	0.028377	2.05	0.003742	2.55	0.000311	3.05	0.00001608
0.06	0.932378	0.56	0.428384	1.06	0.133856	1.56	0.027372	2.06	0.003577	2.56	0.000294	3.06	0.00001508
0.07	0.921142	0.57	0.420184	1.07	0.130227	1.57	0.026397	2.07	0.003418	2.57	0.000278	3.07	0.00001414
0.08	0.909922	0.58	0.412077	1.08	0.126674	1.58	0.025453	2.08	0.003266	2.58	0.000264	3.08	0.00001326
0.09	0.898719	0.59	0.404064	1.09	0.123197	1.59	0.024538	2.09	0.003120	2.59	0.000249	3.09	0.00001243
0.1	0.887537	0.6	0.396144	1.1	0.119795	1.6	0.023652	2.1	0.002979	2.6	0.000236	3.1	0.00001165
0.11	0.876377	0.61	0.388319	1.11	0.116467	1.61	0.022793	2.11	0.002845	2.61	0.000223	3.11	0.00001092
0.12	0.865242	0.62	0.380589	1.12	0.113212	1.62	0.021962	2.12	0.002716	2.62	0.000211	3.12	0.00001023
0.13	0.854133	0.63	0.372954	1.13	0.110029	1.63	0.021157	2.13	0.002593	2.63	0.000200	3.13	0.00000958
0.14	0.843053	0.64	0.365414	1.14	0.106918	1.64	0.020378	2.14	0.002475	2.64	0.000189	3.14	0.00000897
0.15	0.832004	0.65	0.357971	1.15	0.103876	1.65	0.019624	2.15	0.002361	2.65	0.000178	3.15	0.00000840
0.16	0.820988	0.66	0.350623	1.16	0.100904	1.66	0.018895	2.16	0.002253	2.66	0.000169	3.16	0.00000786
0.17	0.810008	0.67	0.343372	1.17	0.098000	1.67	0.018190	2.17	0.002149	2.67	0.000159	3.17	0.00000736
0.18	0.799064	0.68	0.336218	1.18	0.095163	1.68	0.017507	2.18	0.002049	2.68	0.000151	3.18	0.00000689
0.19	0.788160	0.69	0.329160	1.19	0.092392	1.69	0.016847	2.19	0.001954	2.69	0.000142	3.19	0.00000644
0.2	0.777297	0.7	0.322199	1.2	0.089686	1.7	0.016210	2.2	0.001863	2.7	0.000134	3.2	0.00000603
0.21	0.766478	0.71	0.315335	1.21	0.087045	1.71	0.015593	2.21	0.001776	2.71	0.000127	3.21	0.00000564
0.22	0.755704	0.72	0.308567	1.22	0.084466	1.72	0.014997	2.22	0.001692	2.72	0.000120	3.22	0.00000527
0.23	0.744977	0.73	0.301896	1.23	0.081950	1.73	0.014422	2.23	0.001612	2.73	0.000113	3.23	0.00000493
0.24	0.734300	0.74	0.295322	1.24	0.079495	1.74	0.013865	2.24	0.001536	2.74	0.000107	3.24	0.00000460
0.25	0.723674	0.75	0.288845	1.25	0.077100	1.75	0.013328	2.25	0.001463	2.75	0.000101	3.25	0.00000430
0.26	0.713100	0.76	0.282463	1.26	0.074764	1.76	0.012810	2.26	0.001393	2.76	0.000095	3.26	0.00000402
0.27	0.702582	0.77	0.276179	1.27	0.072486	1.77	0.012309	2.27	0.001326	2.77	0.000090	3.27	0.00000376
0.28	0.692120	0.78	0.269990	1.28	0.070266	1.78	0.011826	2.28	0.001262	2.78	0.000084	3.28	0.00000351
0.29	0.681717	0.79	0.263897	1.29	0.068101	1.79	0.011359	2.29	0.001201	2.79	0.000080	3.29	0.00000328
0.3	0.671373	0.8	0.257899	1.3	0.065992	1.8	0.010909	2.3	0.001143	2.8	0.000075	3.3	0.00000306
0.31	0.661092	0.81	0.251997	1.31	0.063937	1.81	0.010475	2.31	0.001088	2.81	0.000071	3.31	0.00000285
0.32	0.650874	0.82	0.246189	1.32	0.061935	1.82	0.010057	2.32	0.001034	2.82	0.000067	3.32	0.00000266
0.33	0.640721	0.83	0.240476	1.33	0.059985	1.83	0.009653	2.33	0.000984	2.83	0.000063	3.33	0.00000249
0.34	0.630635	0.84	0.234857	1.34	0.058086	1.84	0.009264	2.34	0.000935	2.84	0.000059	3.34	0.00000232
0.35	0.620618	0.85	0.229332	1.35	0.056238	1.85	0.008889	2.35	0.000889	2.85	0.000056	3.35	0.00000216
0.36	0.610670	0.86	0.223900	1.36	0.054439	1.86	0.008528	2.36	0.000845	2.86	0.000052	3.36	0.00000202
0.37	0.600794	0.87	0.218560	1.37	0.052688	1.87	0.008179	2.37	0.000803	2.87	0.000049	3.37	0.00000188
0.38	0.590991	0.88	0.213313	1.38	0.050984	1.88	0.007844	2.38	0.000763	2.88	0.000046	3.38	0.00000175
0.39	0.581261	0.89	0.208157	1.39	0.049327	1.89	0.007521	2.39	0.000725	2.89	0.000044	3.39	0.00000163
0.4	0.571608	0.9	0.203092	1.4	0.047715	1.9	0.007210	2.4	0.000689	2.9	0.000041	3.4	0.00000152
0.41	0.562031	0.91	0.198117	1.41	0.046148	1.91	0.006910	2.41	0.000654	2.91	0.000039	3.41	0.00000142
0.42	0.552532	0.92	0.193232	1.42	0.044624	1.92	0.006622	2.42	0.000621	2.92	0.000036	3.42	0.00000132
0.43	0.543113	0.93	0.188437	1.43	0.043143	1.93	0.006344	2.43	0.000589	2.93	0.000034	3.43	0.00000123
0.44	0.533775	0.94	0.183729	1.44	0.041703	1.94	0.006077	2.44	0.000559	2.94	0.000032	3.44	0.00000115
0.45	0.524518	0.95	0.179109	1.45	0.040305	1.95	0.005821	2.45	0.000531	2.95	0.000030	3.45	0.00000107
0.46	0.515345	0.96	0.174576	1.46	0.038946	1.96	0.005574	2.46	0.000503	2.96	0.000028	3.46	0.00000099
0.47	0.506255	0.97	0.170130	1.47	0.037627	1.97	0.005336	2.47	0.000477	2.97	0.000027	3.47	0.00000092
0.48	0.497250	0.98	0.165769	1.48	0.036346	1.98	0.005108	2.48	0.000453	2.98	0.000025	3.48	0.00000086
0.49	0.488332	0.99	0.161492	1.49	0.035102	1.99	0.004889	2.49	0.000429	2.99	0.000024	3.49	0.00000080